

SD Card File Structure

Initial

```
SD Card (Root)
├── chart.js                # JavaScript library used for chart data visualization
```

After First Run

```
SD Card (Root)
├── chart.js                # JavaScript library used for chart data visualization
├── calibration_metadata.csv # Calibrated reference standards
├── wifi_config.csv         # Wi-Fi credentials (SSID/Password), if configured
├── logs/                  # Directory for system data logging
│   ├── s_[date]_[time].csv # Sleeping data log (when synced with internet time)
│   └── s_uptime_[num].csv  # Fallback log (when internet/standard time unavailable)
```

.csv File Data Structure:

s_[date]_[time].csv

Header Row

Time, State, Posture, Mass, TempL, TempC, TempR, SnoreCount, P0, P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, CalibrationID

Field Descriptions

- **Time:** The timestamp of the log entry. It uses real-world time (YYYY-MM-DD HH:MM:SS) if connected to Wi-Fi/NTP, or falls back to system uptime (e.g., Uptime_120s) if offline.
- **State:** The current sleeping state of the user, recorded as either Sleeping or Empty.
- **Posture:** The classified sleeping position determined by the system (Left Side, Right Side, Center Back, Bed Empty, or Needs Calibrating).
- **Mass:** The total pressure force value. In Raw Mode, this represents the sum of all raw sensor values. In Calibrated Mode, it displays the net value (Raw Sum – Noise Sum).

- **TempL / TempC / TempR:** The 5-second moving average temperatures (in °C) captured from the left, center, and right thermistors respectively.
- **SnoreCount:** The running tally of detected snoring events accumulated within the current logging minute.
- **P0 to P11:** The 12 individual raw pressure sensor readings mapped across the 3*4 physical grid layout of the pillow.
- **CalibrationID:** The specific numerical profile identifier used to evaluate the metrics of this log entry, tying it directly to a unique row in the metadata registry.

calibration_metadata.csv

Header Structure (Implicit Data Format)

```
calibration_id,start_time,noise_threshold(12
fields),left_calibration(12 fields),center_calibration(12
fields),right_calibration(12 fields)
```

(Note: This file does not write a text header line to the SD card; it saves snapshots directly as comma-separated values matching this exact column sequence).

Field Descriptions

- **calibration_id:** A unique, auto-incrementing integer key used to version and track configuration changes every time a calibration step or a full system reset occurs.
- **start_time:** The timestamp indicating exactly when that specific configuration snapshot was written to the SD card.
- **noise_threshold (Next 12 fields):** The sequential baseline noise values recorded for each cell in the 3*4 grid during an empty bed calibration. Defaults to 0 if empty.
- **left_calibration (Next 12 fields):** The benchmark target pressure values recorded for each cell in the matrix while calibrating the user's side-sleeping position on the left.
- **center_calibration (Next 12 fields):** The benchmark target pressure values recorded for each cell in the matrix while calibrating the user's back-sleeping position in the center.
- **right_calibration (Final 12 fields):** The benchmark target pressure values recorded for each cell in the matrix while calibrating the user's side-sleeping position on the right.